

Suggested Dissertation Structure

MSc Applied Social Data Science

This document outlines a recommended structure for your dissertation based on your research proposal and literature review on the demographic transition fitness paradox. The structure follows a standard empirical social science dissertation format commonly used in MSc programmes.

1. Title Page

- Dissertation title
- Your name
- Programme and institution
- Supervisor
- Submission date

2. Abstract (250–400 words)

- Research question
- Theory and motivation
- Data and methods
- Main findings
- Contribution

3. Acknowledgements (Optional)

- Short thank-you section

4. Table of Contents

- Table of contents
- List of tables
- List of figures
- Abbreviations

Chapter 1 — Introduction (~2,000–3,000 words)

- Background and motivation
- Research problem
- Research questions
- Hypotheses

- Contribution
- Dissertation structure overview

Chapter 2 — Literature Review (~4,000–6,000 words)

- Demographic transition literature
- Theoretical explanations
- J-curve literature
- Mechanism studies
- Research gaps
- Contribution of the present study

Chapter 3 — Data and Methodology (~3,000–5,000 words)

- Research design
- Datasets and sources
- Variable construction
- Methods for H1
- Methods for H2
- Methods for H3
- Methods for H4
- Robustness checks
- Ethical considerations

Chapter 4 — Descriptive Results (~2,000–3,000 words)

- Cohort fertility trends
- Cross-country patterns
- Education distributions
- Descriptive visualisations
- Preliminary evidence for inversion

Chapter 5 — Main Results: SES-Fertility Gradient and J-Curve (~3,000–5,000 words)

- Fixed-effects results
- Cross-country variation
- Quantile regression
- GAM / spline analysis
- J-curve evidence

- Robustness checks

Chapter 6 — Mechanisms Results (~3,000–5,000 words)

- Machine learning feature selection
- SHAP interpretation
- Bayesian hierarchical modelling
- Mediation analysis
- Cultural vs economic mechanisms

Chapter 7 — Discussion (~2,000–4,000 words)

- Summary of findings
- Theoretical implications
- Comparison with literature
- Strengths and limitations
- Future research directions

Chapter 8 — Conclusion (~1,000–2,000 words)

- Restate research question
- Summarise findings
- State contribution
- Final theoretical implications

References / Bibliography

- Use consistent citation style
- Recommended tools: Zotero, Mendeley, BibTeX

Appendices

- Robustness tables
- Additional figures
- Variable definitions
- Model specifications
- Code and preprocessing pipeline
- SHAP plots and diagnostics
- Supplementary analyses

Final Advice:

Keep the literature review focused on theory and prior research, the methodology chapter focused on empirical strategy and data, and the results chapters focused on findings. This separation makes the dissertation easier to read, more professional, and easier to examine.